

End-to-end Cloud Services & AI-Driven Product Suites for Healthcare, Retail, Logistics, and Automotive



Dear Learner,

Welcome to **Hands-On Knowledge Graph and Graph AI Workshop** from **Compute Elements Saranga Technologies!**

Following are the set of Recommended Pre-requisites to follow along our session.

You are encouraged to complete these installations before our workshop. If you would like to try the exercises as we go through the workshop, these installations will be handy for you.

List of Pre-Requisites

- 1) [Device – Your own Laptop \(Mandatory\)](#)
- 2) [GitHub User Account](#)
- 3) [Python Installation in your laptop \(from python.org\)](#) - Required to run the agents
- 4) [Visual Studio Code IDE Installed \(from code.visualstudio.com\)](#)
- 5) [Docker for Windows Installation](#) – Required to run Graph DB

Below one is optional requirement needed only if you would like to work in Linux like sessions from windows.

- 6) [GIT for Windows](#)

The Program will require you to install some more packages and dependencies and you will be guided to install them as appropriate during the lessons.

Device – Your Laptop

The program involves using IDE to write code, and compile and run the in local environment or virtual environment in Docker containers. A normal Quad Core, 8 GB RAM, 256GB HDD will suffice. If you can use your PC for Browsing Internet, Taking Video Calls and Word Processing, it must be good for our workshop.

Github User Account

Do you have a Github user account? If yes, that is great, keep them handy, you will need it to access the kits.

If you do not have an account, do logon to github.com and create a user account for yourself..

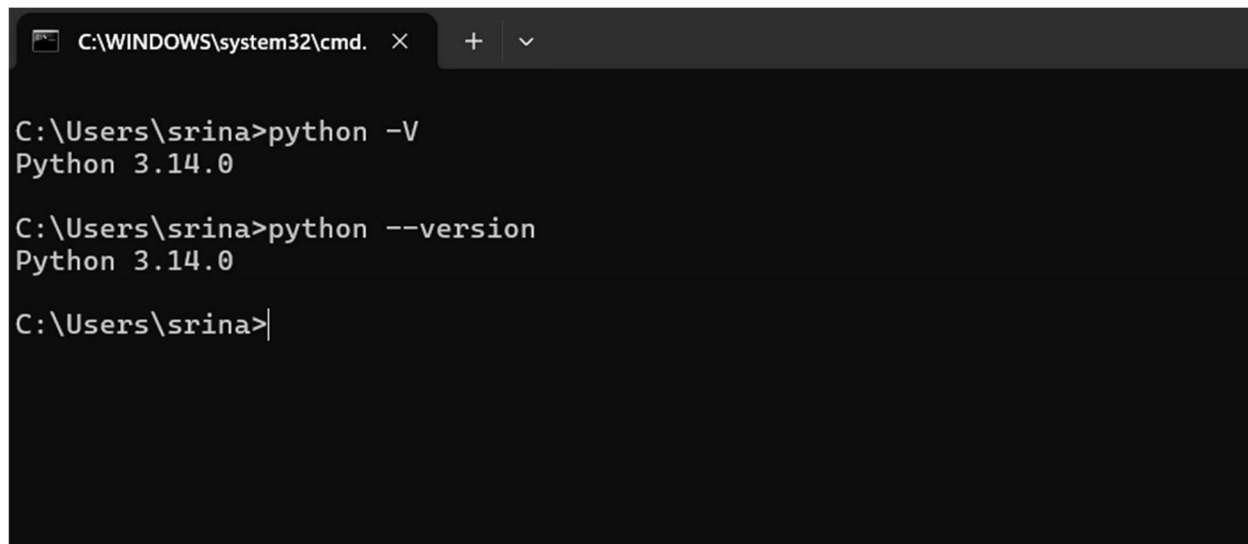
Python Installation

Check if you already have Python installed in your laptop

Open Command Prompt - Press Window Key and in the Run Command box type “cmd” and hit Enter

Once you are in the Command prompt type ***python --version*** (or) ***python -V*** and hit Enter.

Do you see a python version like 3.x as shown below, then you are good.

A screenshot of a Windows Command Prompt window. The title bar shows the path 'C:\WINDOWS\system32\cmd.' and standard window controls. The command prompt shows the user 'srina' at the 'C:\Users\' prompt. The user has entered 'python -V' and 'Python 3.14.0' is displayed. Then, the user enters 'python --version' and 'Python 3.14.0' is displayed again. The prompt is now ready for the next command.

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\srina>python -V
Python 3.14.0
C:\Users\srina>python --version
Python 3.14.0
C:\Users\srina>
```

If you do not have Python installed in your Device or is not 3.x then follow the below instructions

- go to the Python Downloader Page :
<https://www.python.org/downloads/release/python-3142/>
- At the bottom of the page, you will see a link to download the installer for your Operating Systems (macOS or Windows or Source release)
- If you use Windows Laptop, select the “Download Python Installs Manager” for Windows

Files

macOS

Download macOS installer

Windows

Download Python install manager

Source release

Download XZ compressed source tarball

Version	Operating system	Description	File size	Sigstore	SBOM	MD5 checksum
---------	------------------	-------------	-----------	----------	------	--------------

- Once the Installer is downloaded, Run the installer to complete installation of Python in your Laptop.
- Once the installation process completes, you may once again run the ***python -version*** or ***python -V*** command to check the version.

VS Code IDE

Do you have VS Code IDE already installed in your Laptop, if not it is very easy. VS Code IDE is one of the most used IDE. It is very intuitive and learning to code in this will help you build up your Tech stack for future.

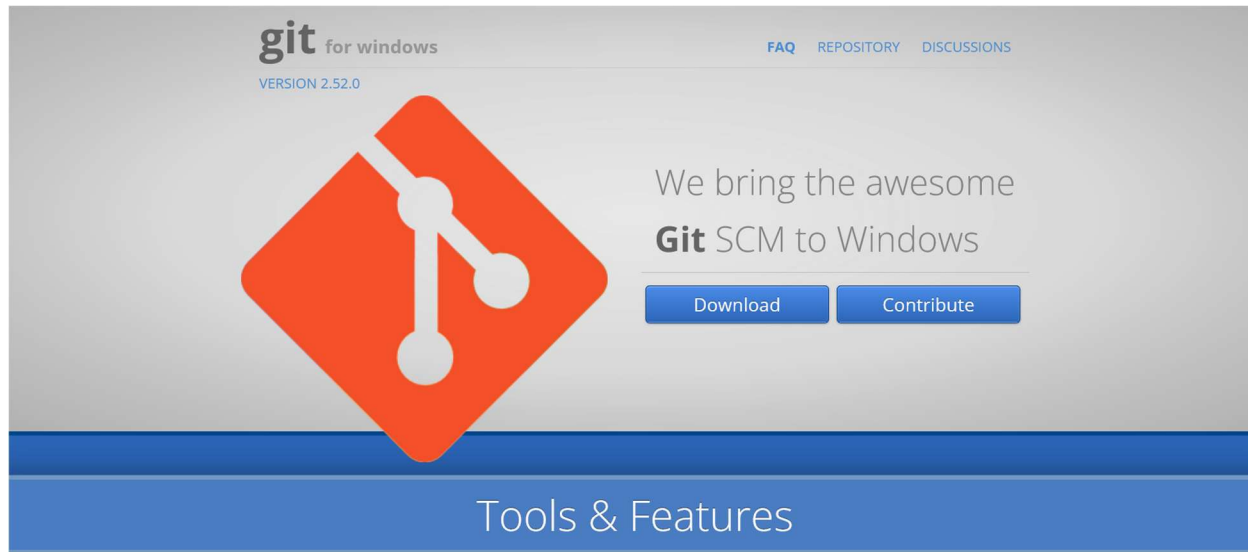
- Visit the page <https://code.visualstudio.com/> and download the package for Windows or your Operating System
- Open the Installer and follow the instructions and prompts to complete the installation.

The screenshot shows the Visual Studio Code website. At the top, there's a navigation bar with links: Visual Studio Code, Docs, Updates, Blog, API, Extensions, MCP, and FAQ. To the right of the navigation bar is a search bar with a magnifying glass icon, a 'Search' button, and a 'Download' button. Below the navigation bar, the main content area features a large, bold text overlay that reads 'The open source AI code editor'. Below this text, there's a 'Download for Windows' button. Further down, there's a link to 'Web, Insiders edition, or other platforms' and a small note: 'By using VS Code, you agree to its license and privacy statements.' At the bottom of the screenshot, there's a preview of the VS Code interface showing a code editor with a file named 'MailList.txt' and a terminal window displaying the command 'import { For, createSignal, createMemo } from "solid-js";'.

GIT for Windows

GIT for Windows provides you with interactive Unix Like terminal for accessing the file system and allows you to run Shell commands for the operations. A very handy tool to practice hands on Unix and Shell scripting, a key necessity in professional setting, and trains you for logging and working on Linux servers which are the most common hosts in a professional deployment.

- Visit <https://gitforwindows.org/>
- Download the Installer
- Run the installer following the instructions and Prompts



Docker For Windows

Docker is a Containerization platform and provides set of tools for Container management, and is one of the most common containerization tool used in Software Development. Instead of spending hours manually installing databases, specific versions of Python, or complex libraries on your computer—and hoping they don't clash with what's already there—Docker lets you download a pre-packaged images for the needs and manages a container, which you can access for your work and at the same time isolates this from other installations in your Laptop.

- Visit <https://docs.docker.com/desktop/setup/install/windows-install/>
- Download the Docker Desktop Version for Windows for your processor architecture
- Run the Installer and Follow instructions and prompts to complete the installation